

Vi starter ud med bare at gå ud fra definitionen af likelihood funktionen tager log likelihood og finder maksimum

$$\begin{aligned}
 \bar{X}_1 &\sim N(\mu, \sigma_0^2), \bar{X}_2 \sim N(\mu, \sigma_0^2), \bar{X}_3 \sim N(2\mu, \sigma_0^2) \\
 L(\mu) &= \prod_{i=1}^3 f_{\bar{X}_i}(x_i) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \cdot e^{-\frac{1}{2\sigma_0^2}(x_1-\mu)^2} \\
 &\cdot e^{-\frac{1}{2\sigma_0^2}(x_2-\mu)^2} \cdot e^{-\frac{1}{2\sigma_0^2}(x_3-2\mu)^2} \\
 &= \left(\frac{1}{\sqrt{2\pi\sigma_0^2}}\right)^3 \cdot e^{-\frac{1}{2\sigma_0^2}((x_1-\mu)^2 + (x_2-\mu)^2 + (x_3-2\mu)^2)} \\
 \ln L(\mu) &= -\frac{3}{2} \ln(2\pi\sigma_0^2) - \frac{1}{2\sigma_0^2}((x_1-\mu)^2 + (x_2-\mu)^2 + (x_3-2\mu)^2) \\
 \frac{\partial}{\partial \mu} \ln L(\mu) &= \frac{1}{\sigma_0^2} \cdot (x_1 - 2\mu + x_2 - 2\mu + 4x_3 - 8\mu) = 0 \\
 \Leftrightarrow 2x_1 + 2x_2 + 4x_3 &= 12\mu \Leftrightarrow \hat{\mu} = \frac{x_1 + x_2 + 2x_3}{6}
 \end{aligned}$$

Vi kan også lige notere at hvis vi differentierer igen mht. μ får $-\frac{6}{\sigma_0^2} < 0$

Da har vi et reelt maksimum

Herefter ved vi at en lineær kombination af normalfordelte variable er normalfordelt altså skal vi bare finde middelværdi og varians

"summen af normal er normal"

$$E[\bar{X}_1 + \bar{X}_2 + 2\bar{X}_3] = E[\bar{X}_1] + E[\bar{X}_2] + 2E[\bar{X}_3]$$

$$= \frac{6\mu}{6} = \mu$$

$$V[\bar{X}_1 + \bar{X}_2 + 2\bar{X}_3] = \frac{1}{6^2} \cdot (V[\bar{X}_1] + V[\bar{X}_2] + 4V[\bar{X}_3])$$

$$= \frac{1}{6^2} \cdot (\sigma_0^2 + \sigma_0^2 + 4\sigma_0^2) = \frac{\sigma_0^2}{6}$$

da $\frac{\bar{X}_1 + \bar{X}_2 + 2\bar{X}_3}{6} \sim N(\mu, \frac{\sigma_0^2}{6})$

Herefter transformere vi variablen så den er standardnormal og udleder 95% konfidensintervallet

Vinormalisere

$$\frac{\bar{X}_1 + \bar{X}_2 + 2\bar{X}_3}{6} \sim N(\mu, \frac{\sigma_0^2}{6})$$

$$\frac{\frac{\bar{X}_1 + \bar{X}_2 + 2\bar{X}_3}{6} - \mu}{\sqrt{\frac{\sigma_0^2}{6}}} \sim N(0, 1)$$

0.95

$$P(-U_{0.975} \leq \frac{\bar{X}_1 + \bar{X}_2 + 2\bar{X}_3}{6} - \mu \leq U_{0.975})$$

$$= P(-U_{0.975} \cdot \sqrt{\frac{\sigma_0^2}{6}} - \mu \leq \bar{X}_1 + \bar{X}_2 + 2\bar{X}_3 \leq U_{0.975} \cdot \sqrt{\frac{\sigma_0^2}{6}} - \mu)$$

$$= P(\bar{X}_1 + \bar{X}_2 + 2\bar{X}_3 \leq \mu + U_{0.975} \cdot \sqrt{\frac{\sigma_0^2}{6}})$$